

Conformal Coating for PCB-Based Electronic Circuits

Coating materials protect electronic assemblies against malfunctions or failures in robust operating conditions. Here is what kind of protection these materials provide, properties of coating materials and why they are in demand

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Before electronic circuits based on printed circuit boards (PCBs) found their advent in industrial applications, electrical conductor wires were protected against short-circuits by applying a coating based on organic resin. With the use of PCB for connecting different components together and development of circuits capable of operating even under extreme conditions, such as in tropical climates and the military, space and automotive sectors, the requirements of protective properties for electronic circuitry have increased.

Polyparaxylylene was employed in the early stages of electronic circuitry as a protective coating in military applications. Its use in other industrial mass production failed due to the high cost of the material, application costs, and especially the costs for masking (covering of areas that should remain uncoated).

As an alternative, inexpensive

0	10	20	30	40	50	60	70	80	90	100
Low humidity				Mean humidity				High humidity		Max. humidity
ESD danger				Smooth operation				Critical operation		Special case

Fig. 1: Influence of relative humidity from 0 to 100 per cent on the operation of electronic assemblies

materials modelled after wire coating were sought for mass production, keeping their coating costs and protective properties in mind.

Therefore coating materials like 'Tempomatlack' were created in the mid 1970s, which were based on oxidation-cured polyurethane resins. This material was suitable, due to its viscosity for both dip coating as well as manual application with a brush. In the course of development, coating materials based on epoxy resins, acrylic resins and silicones were added, which, in various modifications, also met requirements for development of coating technologies such as spray coating and selective coating.

The coating materials also had to be adapted to the latest additions in drying/curing processes. The demand for reduction in emissions led to the emergence of coating materials with reduced organic solvents in view of the volatile organic compounds (VOC) guideline, water-borne lacquers and solvent-free UV hybrid lacquers.

Why are electronic assemblies provided with a coating material

Besides the solder joints recessed by

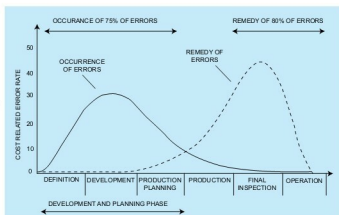


Fig. 2: Occurrence and remedy of errors during product life-cycle

the solder resist, there are also various potentials in contact with the different components on the PCB that are not shielded. Depending on the requirements, conformal coatings, thick-film coatings or casting compounds can be employed to shield against such potentials. Their main task is protection of the completed assembly against malfunctions or failures in robust operating conditions.

For this purpose, the Institute for Interconnecting and Packaging Electronic Circuits (IPC) Standard, under IPC-T-50 terms and definitions, defines a conformal coating as being "an insulating and surface-compliant coating which represents a protective coating against the detrimental influences of the environmental conditions..."

For market success of any electronic product, reliability and customer satisfaction play decisive roles. Therefore manufacturers take necessary

